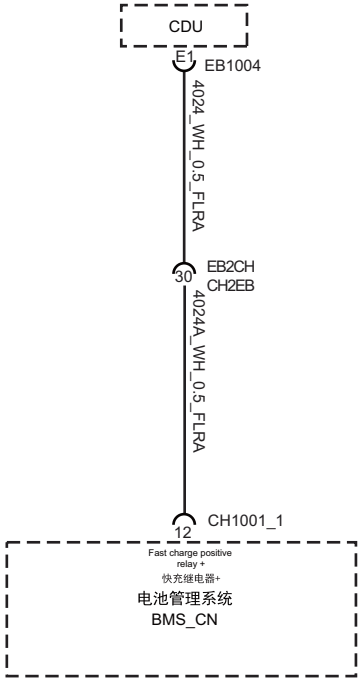
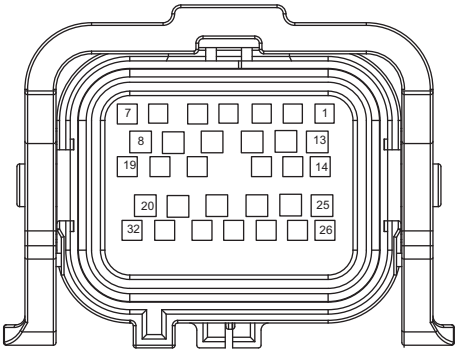
P1B23-00、P1B1F-F0、P1B1F-F1、P1B1F-F2、P1B20-00

Fault diagnosis code
P1B23-00：DC charging positive relay adhesion failure
P1B1F-F0：DC charge positive contactor coil circuit short to GND fault
P1B1F-F1：DC charge positive contactor coil circuit short to power fault
P1B1F-F2：DC charge positive contactor coil circuit open fault
P1B20-00：DC charge positive contactor coil circuit current abnormal fault
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• BMS fault • DC charging positive relay fault • BMS DC charging positive relay circuit fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1B23-00
P1B1F-F0
P1B1F-F1
P1B1F-F2
P1B20-00
To set the effect of a fault code condition
Description of circuit diagram

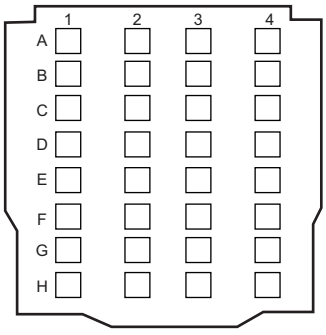
Circuit diagram



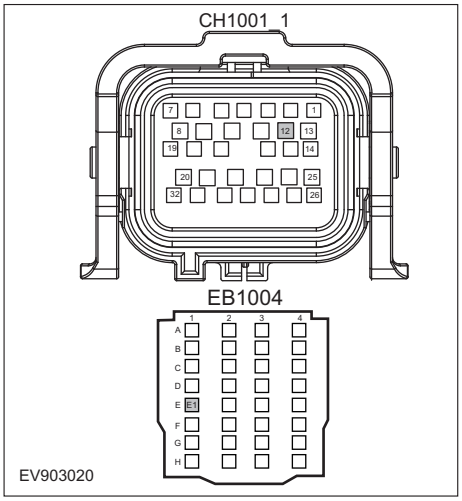
CH1001_1

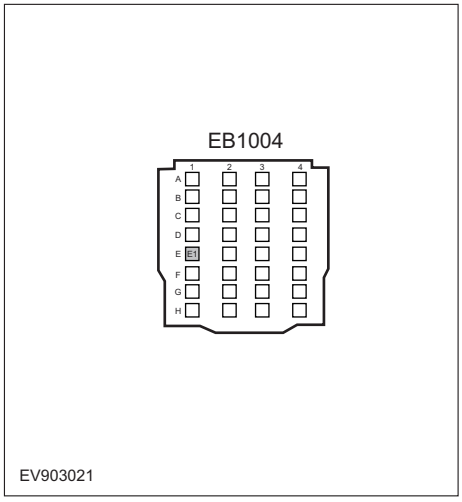


EB1004



EV903019

Troubleshooting steps	
Step 1	General inspection.
	1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 Is there any other fault code except for P1B23-00/P1B1F-F0/P1B1F-F1/P1B1F-F2/P1B20-00?
	YES → Refer to:DTC Summary list(BMS).
NO ↓	
Step 3	Check the circuit between BMS and CDU. (Open circuit)
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug CH1000_1 of BMS. 3 Disconnect the harness plug EB1004 of CDU. 4 Measure the resistance between the harness terminals of BMS and CDU. Measuring circuit: resistance between terminal 12 on plug CH1000_1 and terminal E1 on plug EB1004. **Standard value: < 1Ω** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 4	Check CDU circuit. (Short circuit)
	1 Turn the ignition switch to OFF. 2 Connect the harness plug CH1000_1 of BMS. 3 Measure the resistance between the harness terminal of CDU and ground. Measuring circuit: resistance between terminal E1 on plug EB1004 and ground. **Standard value: ∞** 4 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal E1 on plug EB1004 and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the connecting cable of BMS.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Check the connecting cable of CDU.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of CDU. 5 Check whether the CDU connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.

YES ↓		
Step 7	Check the DC charging positive relay circuit.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the CDU end cover. 4 Check whether the main positive relay circuit exist the situation of open or short circuit. 5 Check whether the main positive relay circuit exist the situation of shorting to battery. 6 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 8	Check DC charging positive relay signal checking module.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the CDU end cover. 4 Check whether the DC positive relay signal checking module is damaged. 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 9	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
		NO → Repair or change the BMS module.
YES ↓		

Step 10	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1000_1 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	